

GENERAL CHEMISTRY DRY LABS USING AVOGADRO



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General Chemistry Dry Labs Using Avogadro

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Licensing

A detailed breakdown of this resource's licensing can be found in [Back Matter/Detailed Licensing](#).

SECTION OVERVIEW

1: A User's Guide to Avogadro

This is an introduction to Avogadro built as an introduction for the following General Chemistry dry labs. A complete manual can be found at <https://avogadro.cc/docs/getting-sta.../introduction/>

1.1: Basic Tools

1.2: Building Your Molecules - Energy Optimization

1.3: Generating an Electrostatic Potential Map of a Molecule

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1.1: Basic Tools

- 1) Avogadro has built in molecules. To access them, go to the toolbar under **Build > Insert > Fragment**.

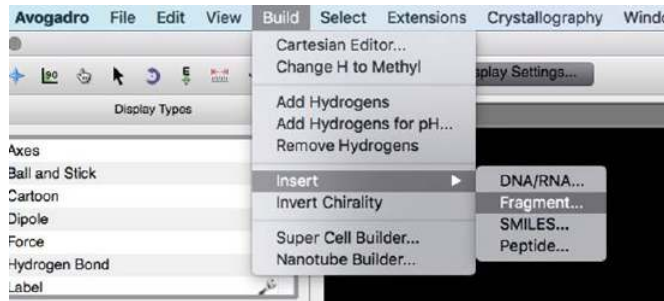


Figure 1.1.1: Copy and Paste Caption here. (Copyright; author via source)

There are various built in molecules and compounds as shown. We are going to start with the simplest but one of the most essential molecule known to mankind H₂O! Click on water at the bottom.

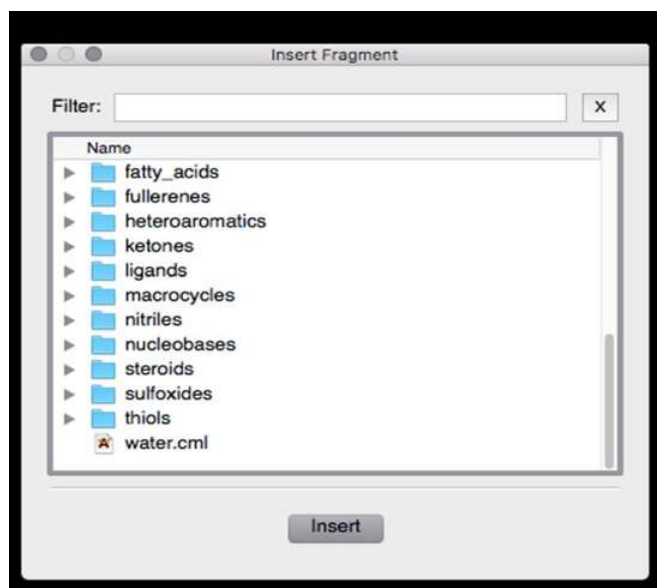


Figure 1.1.2: List of prebuilt molecules. (Copyright; author via source)

- 2) The molecule should appear as follow, see Figure 1.1.3. The way it may actual appear on your computer varies by what is defaulted by Avogadro when you downloaded it.

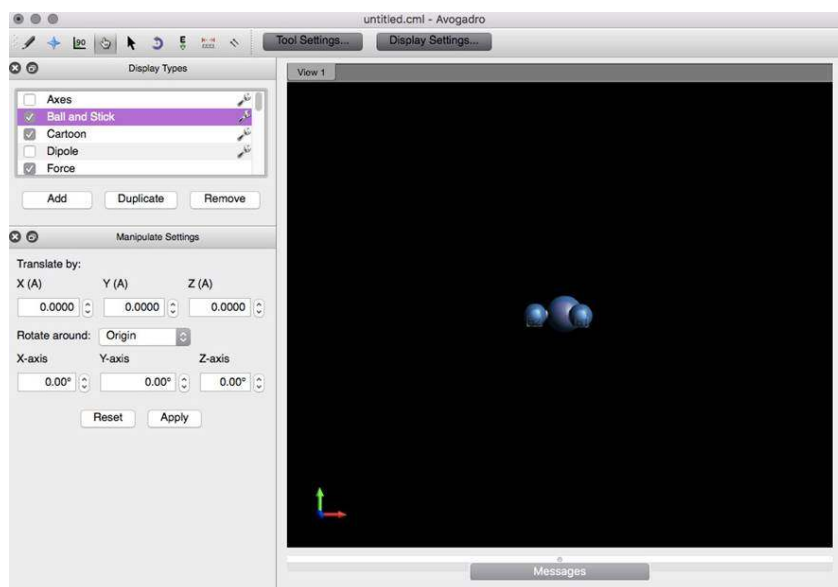


Figure 1.1.3 : Water molecule(Copyright; author via source)

3) Because the software is 3D representation of the molecule (xyz), you will have to adjust the position of the molecule to your liking. To do so, select the diamond star button on the ribbon above the display types, see fFigure 1.1.4. You can rotate the molecule about an atom by clicking on the atom, holding the left mouse button, and moving your mouse as needed.

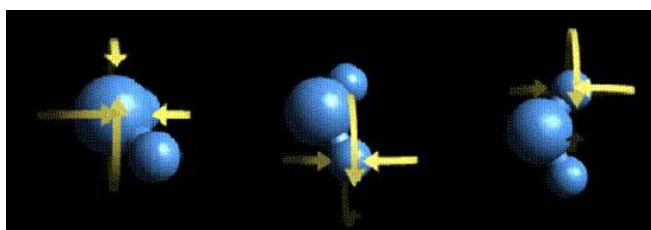
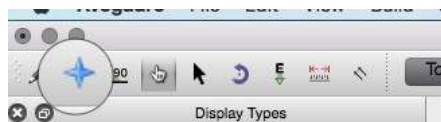


Figure 1.1.4: Use the diamond star button for rotation and rotate the center oxygen atom of one of the two outer hydrogen atoms. (Copyright; author via source)

4) As shown in Figure 1.1.4, the whole molecule, that is each individual bond and atom, is highlighted in blue. To unselect all the atoms, click the **black arrow** button on the ribbon above the **Display Type** and right click on the black space next to the molecule. If you want to reselect a specific bond or atom select the **black arrow** button and then click on the desired atom or bond, see figure V. You can select multiple atoms or bonds by holding the CLRT button and clicking on the desired atoms or bonds.

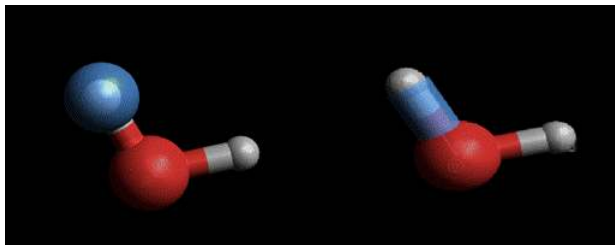


Figure 1.1.5: Use the arrow button for highlighting atoms or bonds. Here is an example with a highlighted atom on the left figure and highlighted bond on the right figure(Copyright; author via source)

5) To view the bond length between atoms, at the top toolbar, go to **View > Properties > Bond Properties**. The bond properties will show as below: You can highlight the bond number to know which bond is what. For example, if you wish to know which bond is “Bond 1” and which atoms are forming that bond you simply click on **Bond 1** and the bond will then be highlighted in blue. You could also label the atoms by simply selecting the **Label** check box under the display type box for better visuals. You can also measure bonds by using the **Ruler Button** and selecting the atoms. Overall you can explore other properties such as **Atom Properties** or **Angle Properties** similarly. See figure VI.

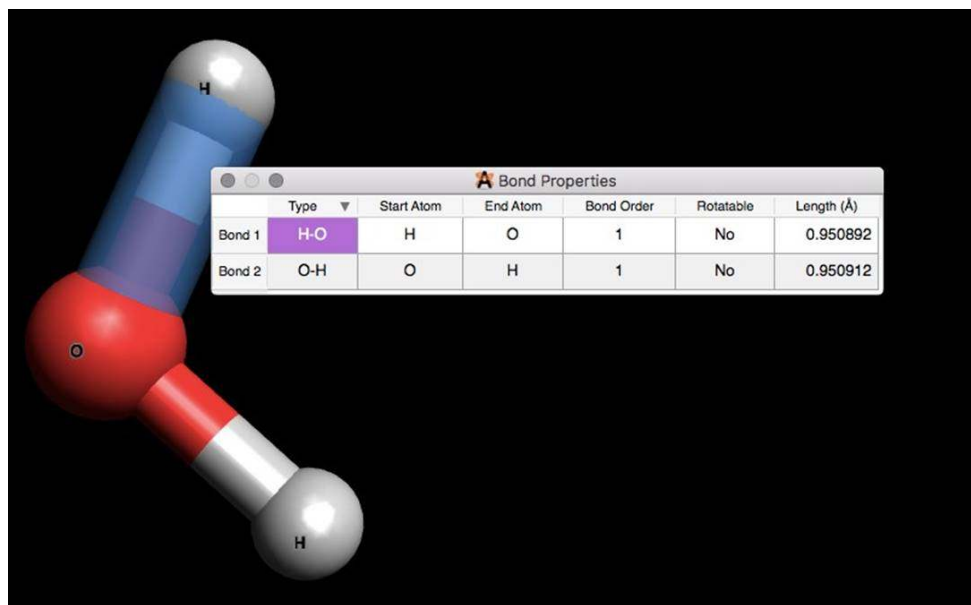


Figure 1.1.6: Bond properties after selecting “Bond 1” (Copyright; author via source)

6) This is generally the basics as far as handling molecules. We will now continue with on another tool before going into building your own specific molecules. There is the **Manipulation Tool** (the hand button). The button and functions are shown in Figure 1.1.7. You can “stretch” atoms and bonds with this tool (could be a hassle if you need to build your molecules with a specific bond length and could ruin it if you are not careful). So, please be weary when using this tool. I would suggest just sticking to the **black arrow** when working with the molecules. Remember though you can always start over

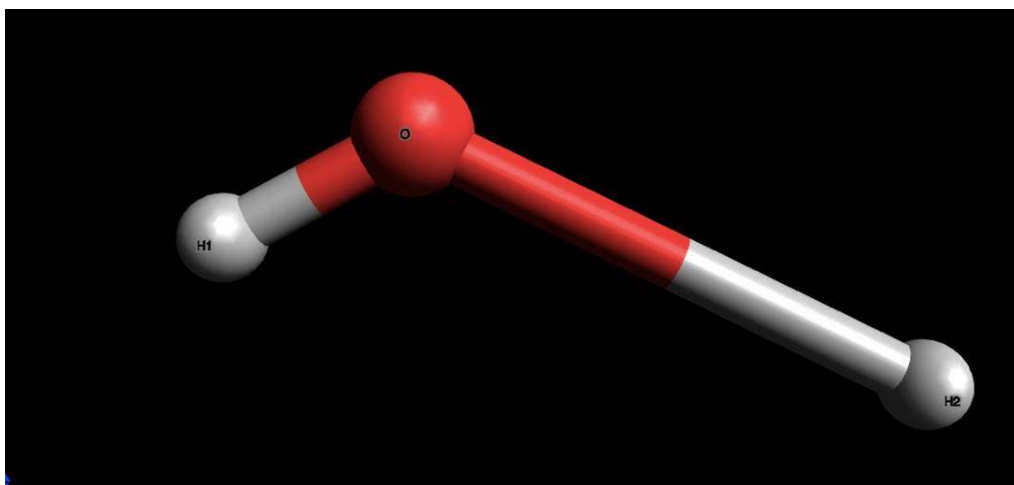
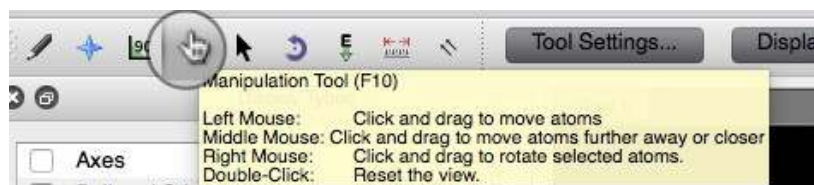


Figure 1.1.1: The “Hand button” or manipulation tool with functions is circled. Pulling on the OH bond stretches it. You would not want your molecule to look like this.(Copyright; author via source)

Note: You could fix the stretched water molecule with a method that is introduced in section two.

7) However, the hand button is useful when you are trying to shift the position of the **WHOLE MOLECULE**. Simply highlight the whole molecule blue (see figure III and step 4), click any part of the highlighted molecule and shift to the position you desire.

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1.2: Building Your Molecules - Energy Optimization

Before we proceed into building molecules, I would like to first briefly introduce **Energy Optimization** or sometimes known as **Energy Minimization**. Avogadro finds the most stable form, lowest energy, of the molecule by minimizing all the interactions within the molecule. It does so by using advanced algorithms by applying a force field.

1) We are going to begin building Methane (CH_4). Click the pen, or **draw tool**, on the tool ribbon as shown in Figure 1.2.1 below.



Figure 1.2.1: The drawing tool is circled (Copyright; author via source)

2) The **display types** window and **tool settings** window should be on your screen. If not, simply click on the **tool settings** and **display settings** next to the tool ribbon. From the draw settings window, you are able to select the elements you wish to draw with. There is also an **adjust hydrogens** checkbox. This adds hydrogens to the atom based off the valency of the atom in order to fulfill the octet rule. For example, click on element drop box in the draw settings window and select carbon, see figure II. On the black screen, click on the spot you wish to put the atom. The carbon comes up with four hydrogen atoms (CH_4), see figure II. If you do this with oxygen, the atom will appear with two hydrogens (try some other atoms on your own and see the octet rule in action).

Note: The actually setting that would be found in the tool setting window depends on which tool you have selected and are working with.

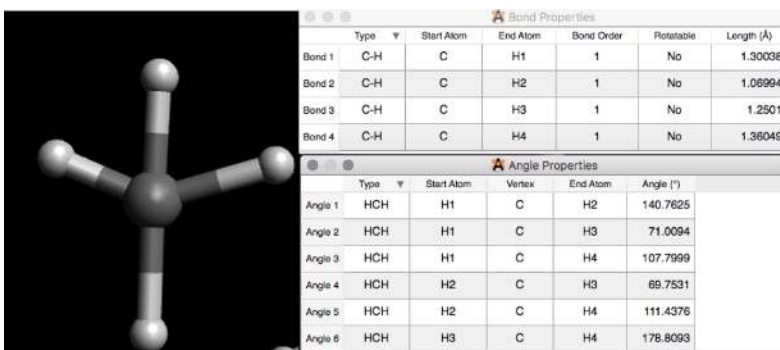


Figure 1.2.2: To draw settings with carbon and adjust hydrogens selected (left). Drawn Methane, CH_4 (right) (Copyright; author via source)

3) CH_4 (methane) has a tetrahedral geometry. The angle and bond distances are as follows:

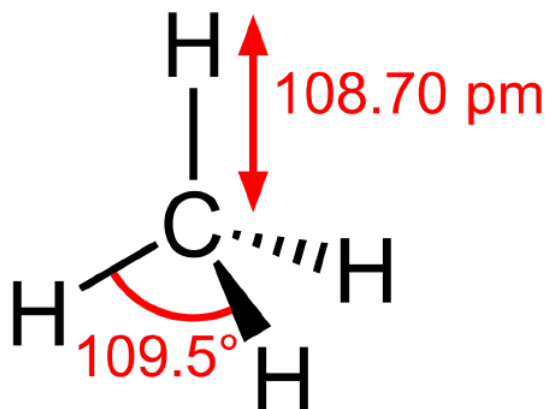


Figure 1.2.3: 3-D Diagram of Methane (Copyright; author via source)

where the distance is given in picometers. Please note Avogadro gives the distance in angstroms. To check the angles, go to **angle properties** (see step 5 in Section 1). To check the lengths, go to **bond properties** (see step 5 in Section 1). Because methane is a simple molecule, and if you have the **adjust hydrogen** checked off, then Avogadro usually gives the correct geometry. See below, figure IV, for the bond and angle properties of the methane drawn in Figure 1.2.2. Geometry is important in energy optimization.

A screenshot of the Avogadro software interface. On the left is a 3D ball-and-stick model of a methane molecule. On the right, two windows are open: 'Bond Properties' and 'Angle Properties'.

	Type	Start Atom	End Atom	Bond Order	Rotatable	Length (Å)
Bond 1	C-H	C	H1	1	No	1.07
Bond 2	C-H	C	H2	1	No	1.07
Bond 3	C-H	C	H3	1	No	1.07
Bond 4	C-H	C	H4	1	No	1.07

	Type	Start Atom	Vertex	End Atom	Angle (°)
Angle 1	HCH	H	C	H	109.4710
Angle 2	HCH	H	C	H	109.4713
Angle 3	HCH	H	C	H	109.4713
Angle 4	HCH	H	C	H	109.4713
Angle 5	HCH	H	C	H	109.4713
Angle 6	HCH	H	C	H	109.4711

Figure 1.2.4: Methane properties (Copyright; author via source)

4) For the purpose of understanding energy optimization, let's draw methane without the **adjust hydrogens**. To do so, click the **draw tool**, select carbon from the element list, and place anywhere on the black screen. Then select hydrogen. In order to make the bond between atoms, you must **CLICK AND HOLD** on the starting atom (carbon), and then **DRAG** from the clicked atom (carbon) over to the position in which you want the other atom (hydrogen) to be. See figure VI for the drawn molecule.

NOTE: If you wish to make a single bond to a double bond, click on the bond once. To make it a triple bond, click on it once more.

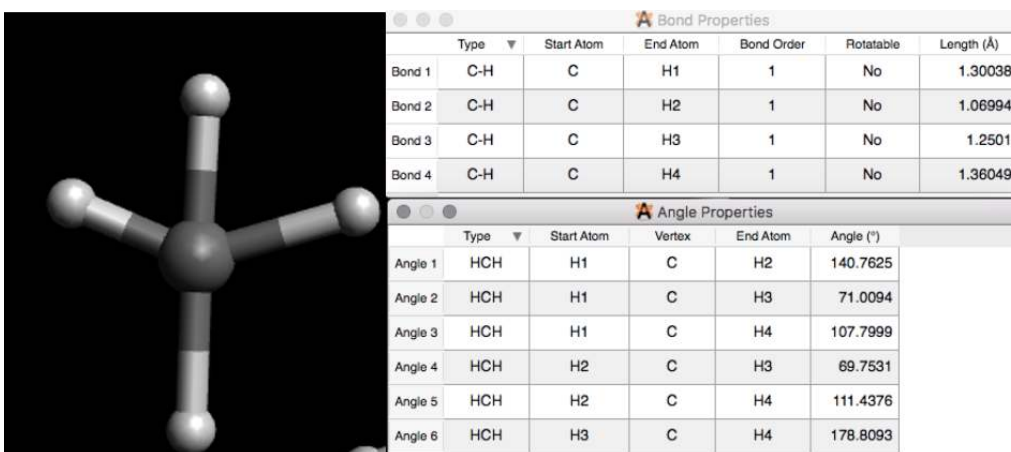


Figure 1.2.5:

Methane drawn without the adjust hydrogens setting (left). Properties of the Methane(Right).(Copyright; author via source)

The length and angles are not in the correct geometry. To fix this, we are going to use the **auto optimization tool**, Figure 1.2.6. Avogadro applies a force field (using some very fancy algorithms of course), to obtain the optimized structure. Working with larger and more complex structures could sometimes lead to using different optimization options. But, for now we will use the settings shown in Figure 1.2.6 for the simple and small molecules we are working with. Click on **Start**. You **Stop** the optimization once the change in energy (dE) is equal to zero or about zero, Figure 1.2.6. The E is equal to the lowest energy value. Check the properties. You should now have the correct geometry.

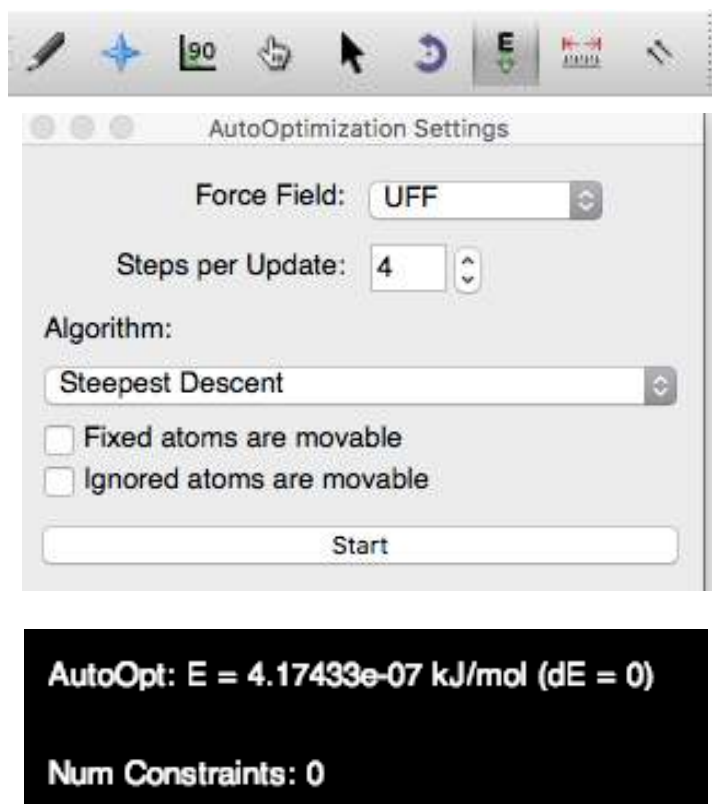


Figure 1.2.6: The auto optimization tool (top). The auto optimization settings that we will be working with (second middle). The dE = 0 stopping point (bottom). (Copyright; author via source)

5) Now let's do Ammonia (NH₃). Ammonia has a trigonal geometry. The lone pair of electrons repel the hydrogen atoms making an angle of 107°. Draw ammonia with the adjusted hydrogens. Check the bond angles. If the bond angles do not equal 107°, then

optimize the structure. Sometimes, you may not get the exact bond angle using the **auto optimization tool**. This is usually not a problem if the angle is almost or about the desired angle. However, you could also get a better geometry through one more step. Go to the toolbar under **Extension > Optimize Geometry**. Check to see if you now have an angle closer to the one you desire. See Figure 1.2.7 for drawn ammonia and the angle properties.

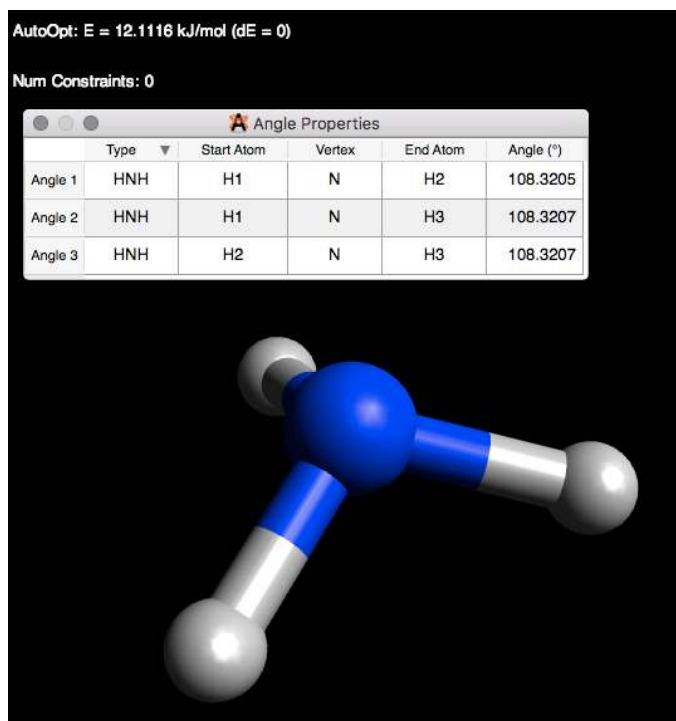


Figure 1.2.7: Copy and Paste Caption here Drawn and Optimized Ammonia. (Copyright; author via source)

6) Now, we will look at an optimized iodomethane (CH_3I). See Figure 1.2.8. Try optimizing the structure on your own!

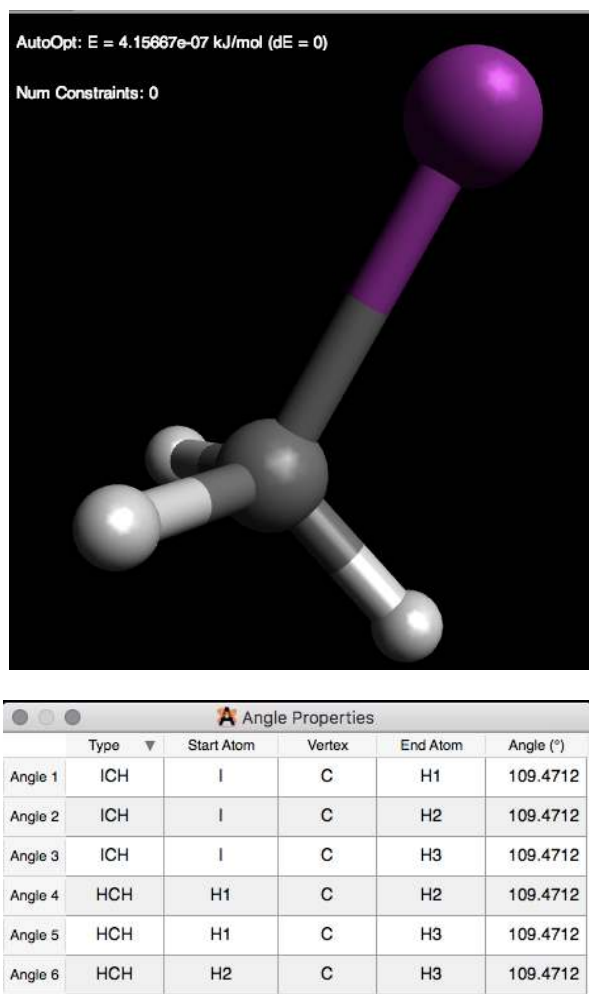


Figure 1.2.8: Optimized iodomethane (top) and properties (bottom). (Copyright; author via source)

Try drawing this structure for fun and practice!

The benzene ring (C_6H_6) is very important organic compound and is classed as an aromatic hydrocarbon. Fun fact: the word “aromatic” comes from the fact that these compounds usually produces a sweet aroma

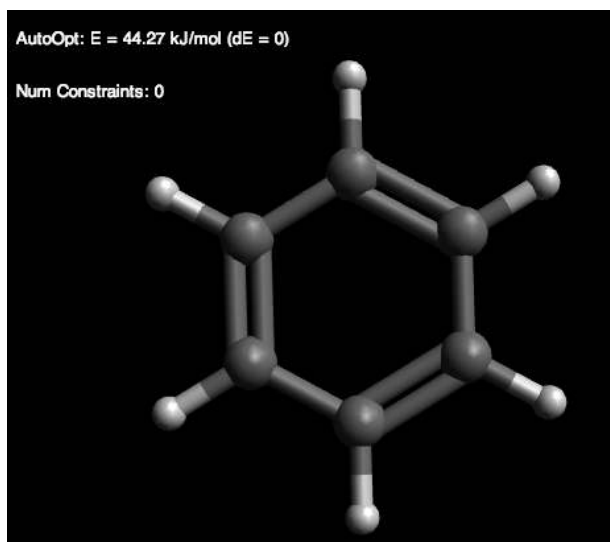


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1.3: Generating an Electrostatic Potential Map of a Molecule

An electrostatic potential (ESP) surface or map of a molecule shows the partial distribution of charge along the molecule's surface. These maps are very useful to help determine molecular polarity and allow for the placement of the dipole arrow.

To generate the map, first the molecular surface must be created. Go to Extensions on the menu bar and select Create Surfaces... at the bottom of the menu.

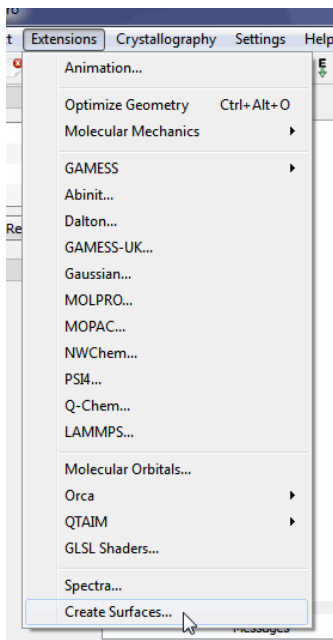


Figure 1.3.1: The Extensions Menu drop down menu. (Copyright; author via source)

The Create Surfaces pop-up menu then appears. Using the dropdown arrow on Color By: select Electrostatic Potential and then press the Calculate button.

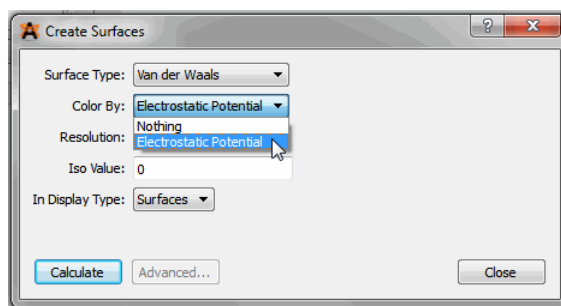


Figure 1.3.2: Create Surfaces pop-up menu (Copyright; author via source)

The ESP surface will be generated, with **red areas being negative** and **blue areas being positive**.

Here is the ESP map for the water molecule. The two lone pairs on the water molecule are slightly negative and the hydrogens are slightly positive. The computer generated dipole arrow is also shown.

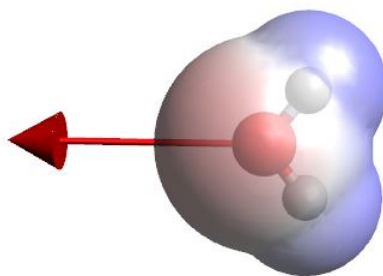


Figure 1.3.3: Water electrostatic potential map showing the electric dipole (Copyright; author via source)

The calculated dipole moment is 1.176 D.

Here are the ESP maps of two dichlorobenzene isomers, which are planar molecules. Are they polar or nonpolar molecules? If polar, draw the dipole arrow. You may want to build these two molecules and generate the ESP maps.

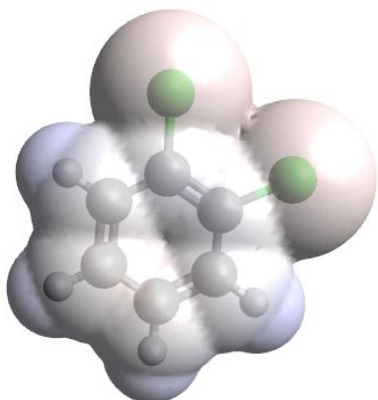


Figure 1.3.4: 1,2-dichlorobenzene (Copyright; author via source)

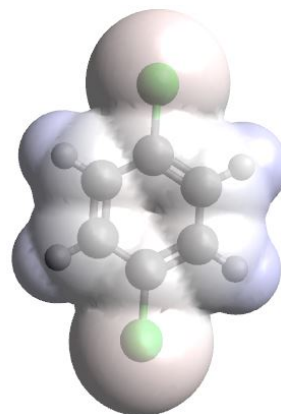


Figure 1.3.5: 1,2-dichlorobenzene. (Copyright; author via source)

Build and produce the ESP maps for ethane (C_2H_6), ethene (C_2H_4), and ethyne (C_2H_2). You may want to do Lewis dot structures for the molecules. How do the three compare to each other?

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2: In class Lab

CHM 2000 Lab Activity Sinex

Investigating Chemical Structures using Avogadro

Download the Avogadro molecular editor at <https://avogadro.cc/>.

Build the following molecules, select optimize geometry, and then perform an energy minimization. For organic-type molecules (carbon-based), use the MMFF94 force field. Insert an image of each molecule from Avogadro using Snipping Tool, a screen capture program in Windows.

Build ethene, $\text{H}_2\text{C}=\text{CH}_2$

What is the geometry of the carbon? What is the H-C-H bond angle?

Is the molecule polar or nonpolar?

Now let's substitute some chlorine atoms for the hydrogen and re-optimize the structures. **Isomers** are compounds with the same molecular formula ($\text{C}_2\text{H}_2\text{Cl}_2$), but the atoms are arranged differently (different connectivity). trans - across or opposite sides cis - same side

1,1-dichloroethene	trans-1,2-dichloroethene	cis-1,2-dichloroethene

Measure the bond angles, Cl-C-Cl in the first one, Cl-C-H in the other two. Record the values.

Calculate the dipole moment and record it for each molecule. Verify the dipole moments by looking them up in the *CRC Handbook of Chemistry and Physics*.

Since the dichloroethene molecules are planar, draw the dipole arrow on them if appropriate.

Go to <http://materia.fisica.unimi.it/manini/dida/structures.html> and retrieve the crystalline structure of graphene.xyz, an unusual two-dimensional or atomic layer solid (right click and Save link as...). Import (under the file menu) the structure into Avogadro. It may take a little time to load. **DO NOT optimize geometry or minimize energy!!!!** Copy and paste the structures below. Change the display type to van der Waals Spheres (electron cloud or space-filled model). Describe the electron cloud produced on the graphene sheet.

Graphene should have alternating single and double carbon-carbon bonds. From the bond lengths, what can you infer?

Graphene has an electrical conductivity comparable to metals such as copper and a strength comparable to steel.

Build a carbyne, which is a chain of carbon atoms with alternating single and triple carbon-carbon bonds. Make your chain with a total of 12 carbon atoms, add the first triple bond starting at the end of the chain, select optimize geometry, and then perform an energy minimization.

What is the geometry of the carbons in the chain?

Measure the bond lengths in your carbyne structure:

triple bond single bond

How do the measured lengths compare to typical length of a C-to-C single and triple bonds?

Carbyne, a 1D network covalent material, may be the strongest material known to humankind!

For more info, see *Route to Carbyne: Scientists Create Ultra-Long 1D Carbon Chains* at <http://www.sci-news.com/othersciences/chemistry/carbyne-ultra-long-1d-carbon-chains-03770.html>.

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3: Project

CHM 2000 Project Sinex

Investigating Chemical Structures using Avogadro II

Download the Avogadro molecular editor at <https://avogadro.cc/>.

Build the following molecules, optimize geometry, and then perform an energy minimization. For organic-type molecules (carbon-based), use the MMFF94 force field. Insert an image of each molecule from Avogadro using Snipping Tool in Windows.

1



Record the bond lengths for the carbon bonds.

Single bond length Double bond length

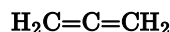
2

Benzene, C_6H_6 where the six carbons form a ring. Do a Lewis dot structure!

Record the bond lengths for the carbon bonds. Are they single or double bonds according to length?

What is the geometry of each carbon atom?

3



Record the bond lengths for the carbon bonds.

How are the hydrogens on the two ends of the molecules arranged (one end compared to other end of the molecule; rotate the molecule to look down the $\text{C}=\text{C}=\text{C}$ axis)?

4

Isomers are compounds with the same molecular formula, but the atoms are arranged differently (different connectivity). Build the two isomers of SCl_4F_2 and do an energy minimization (use the UFF force field). Are the isomers polar or nonpolar. If polar, retrieve the dipole moment.

Copy and paste the structures of the two isomers.

5

Go to <http://materia.fisica.unimi.it/manini/dida/structures.html> and retrieve the crystalline structures of diamond.crystal.xyz and graphite.xyz (right click and Save link as...). Import (under the file menu) the structures into Avogadro. They will take some time to load. Do NOT optimize geometry and perform an energy minimization. Copy and paste the structures below. What is the geometry of the carbon in each structure?

Change the display type to van der Waals Spheres (electron cloud or space-filled model). Which structure has the tighter packing? Why?

What holds the layers in graphite together in the crystalline structure?

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4: Class collaborative activity

CHM 2000 Sinex

Molecular Polarity Collaborative Project

A class collaboration activity for 20 molecules - 4 per student to start and maybe more!

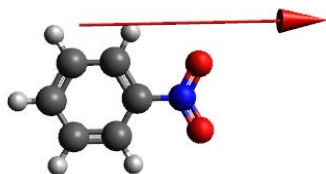
Using a variety of carbon compounds, build the structure, optimize geometry, and then minimize energy in Avogadro. *The golden rule for carbon in organic compounds: C always has four and only four bonds!* For organic-type molecules (carbon-based), use the MMFF94 force field.

Produce each structure and have a fellow student proof it and then do the following:

- Dipole moment - calculated by Avogadro (record the value)
- Dipole moment - measured using NIST data (will be given)
- Dipole arrow
- Screen capture the image using a white background, keep tight around the structure, and then save the file! Windows Snipping tool allows you to save screen captures!

Compile a shared document (spreadsheet supplied by your instructor) with the following information for each of your molecules:

1. Chemical name
2. Chemical line formula
3. Chemical structure (use white background) showing dipole arrow (on menu bar: Insert > Image... to get the file into the spreadsheet) Resize your image to fit in the cell!
4. Prepared by (use your initials)
5. Calculated dipole moment from Avogadro
6. Experimental dipole moment from NIST
7. Dummy column to create $y = x$ line
8. Difference (calculated dipole moment - experimental dipole moment)



Link to spreadsheet that all can edit:

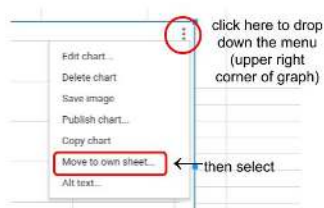
<https://docs.google.com/spreadsheets/d/1FGGe4mvGkQ2e9KSTUhH19TmKZI7klCrA1nwtcRufVN0M/edit?usp=sharing>

Nitrobenzene is given as an example in the spreadsheet and shown here to the right.

Now build and add your assigned molecules to the spreadsheet.

Compare calculated vs. measured dipole moment (divide the work up amongst the class!)

- Plot calculated dipole moment as a function of measured dipole moment
- Add the $y = x$ comparison line (copy calculated dipole moment column into dummy column) Show only regression line through data with no data points!
- Perform a linear regression on the dipole moments data, plot with equation and r-squared value
- Calculate difference (calculated dipole moment - experimental dipole moment)
- On a separate plot, plot the difference as a function of measured dipole moment



After inserting a graph, the graph should be moved to its own tab or sheet. See image to the right.

This should be done for both plots.

How to use the $y = x$ comparison line. See the following Excelets,

“How do you compare data from two different sources?” at http://academic.pgcc.edu/~ssinex/excelets/comparison_plots.xls and/or

“Rulers and Measurement Error”, erroneous ruler II tab at http://academic.pgcc.edu/~ssinex/excelets/Ruler_error.xls

Conclude how good are the Avogadro calculated dipole moment results.

Experimental Dipole Moments: <https://cccbdb.nist.gov/diplistx.asp>

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